

NYC 311 OPERATIONS & SLA ANALYSIS

Executive Summary

SQL analysis and Power BI reporting of public-service request demand, geographic patterns, agency resolution performance, and recorded due-date compliance.

Analysis window

October 11 – November 29, 2025 (50 complete days)

Tools

PostgreSQL · pgAdmin · Power BI · GitHub

Prepared by Muna Mohammed Husen

Data Science Student | Aspiring Data Analyst

PROJECT AT A GLANCE

539,545	529,057	520,123	6.8 hours
Raw records in extract	Requests in main analysis	Valid resolution durations	Median resolution

Executive takeaway. Demand was concentrated in parking, housing conditions, and noise complaints. Resolution performance varied substantially across agencies, and the gap between the 6.8-hour median and 357.4-hour P90 indicates a long tail of requests that took much longer to close. Due-date compliance was assessed separately because recorded due dates were available only for a limited subset of requests.

This summary distinguishes the full extracted file from the main 50-day analysis window and separates general resolution-time benchmarks from recorded due-date compliance.

1. Business Context & Analytical Approach

NYC 311 is a public channel through which residents report service needs and neighborhood issues. This project uses a fixed extract from NYC Open Data to examine demand patterns and resolution performance over a focused period. The goal is to turn raw request records into evidence that can support operational investigation and resource planning.

Business questions

- Which request types generate the most demand?
- How does demand vary by borough and agency, and which problem-location combinations stand out?
- How long do requests take to resolve, and how different are resolution times across agencies?
- Where recorded due dates exist, what share of eligible closed requests met the recorded deadline?

Data and scope

Source: NYC Open Data, 311 Service Requests from 2010 to Present. The extracted file contained approximately 539,545 records. The main analysis uses October 11 through November 29, 2025: 50 complete days. Partial boundary days were excluded so daily demand comparisons use complete days.

Workflow

01 Explore	Inspect counts, date coverage, categories, statuses, and missing values.
02 Investigate	Check timestamp validity, zero-hour durations, missing close dates, and due-date coverage.
03 Prepare	Preserve raw data and create a separate SQL-cleaned analytical table.
04 Analyze	Analyze request demand, borough/problem patterns, resolution time, and due-date compliance.
05 Communicate	Build a Power BI dashboard and validate reported metrics against SQL.

Data quality decisions

The raw table was preserved. 117 records had a close timestamp earlier than the creation timestamp; their resolution duration was set to NULL for resolution analysis, without deleting the original records. Records with no closed timestamp were retained but could not contribute a completed resolution duration. Zero-hour resolutions were retained as valid observations rather than automatically treated as errors.

2. Demand & Geographic Findings

Request demand was concentrated in a small group of recurring service issues. In the 50-day analysis period, the leading categories were Illegal Parking, HEAT/HOT WATER, and Noise - Residential.

Request type	Requests
Illegal Parking	79,984
HEAT/HOT WATER	75,167
Noise - Residential	74,811
Blocked Driveway	25,974
Noise - Street/Sidewalk	19,649
UNSANITARY CONDITION	17,098
PLUMBING	9,982
Noise - Commercial	8,892
Noise	8,636
PAINT/PLASTER	8,365

Demand varied by day

Daily volume varied across the 50 complete days, with lower request counts generally appearing on weekends and a notable drop on Thanksgiving Day (November 27). The highest daily counts included October 31 (13,908 requests), November 12 (13,453), and October 30 (12,946). The lowest included November 27 (8,147) and November 9 (8,187). These patterns describe this extract and period; they should not be generalized to the full year without further analysis.

Location and problem type should be considered together

The strongest borough/problem combinations showed that citywide totals can hide geographically concentrated needs. Examples included Noise - Residential in the Bronx, Illegal Parking in Brooklyn and Queens, and HEAT/HOT WATER in the Bronx, Brooklyn, and Manhattan. A useful operational view should therefore compare both borough and request type, not borough volume alone.

Implication: Prioritization can be more specific when planners identify which problem types dominate in which boroughs, rather than allocating attention using only overall request counts.

3. Resolution Performance

Of 529,057 requests in the main analysis period, 520,123 had a valid resolution duration (approximately 98.3%). The remaining 8,934 records were not deleted: they were excluded only from calculations that require a valid duration, most commonly because the request did not have a usable closed timestamp or had an invalid timestamp sequence.

205.0 h	6.8 h	357.4 h
Average resolution	Median resolution	P90 resolution

Why median and P90 matter

The average of 205.0 hours (about 8.5 days) is much higher than the median of 6.8 hours. This indicates a strongly right-skewed distribution: many requests were resolved relatively quickly, while a smaller group took much longer.

Median: 50% of valid resolution durations were 6.8 hours or less.

P90: 90% were 357.4 hours or less; the slowest 10% took longer than approximately 357.4 hours (about 14.9 days).

The median better describes the middle or typical observation, while P90 helps expose the long tail of slower resolutions. Reporting only the average would hide this difference.

Analytical time benchmarks

Benchmark	Share of validly resolved requests
Resolved within 24 hours	61.2%
Resolved within 72 hours	77.8%
Resolved within 7 days	84.9%

These 24-hour, 72-hour, and 7-day thresholds are analytical benchmarks created for this project, not official NYC 311 service-level agreements.

Agency-level differences

Median resolution times varied widely. Selected results included NYPD at 1.2 hours, DHS at 7.3 hours, DEP at 29.2 hours, DSNY at 34.1 hours, HPD at 54.7 hours, DOT at 72.2 hours, TLC at 2,332.8 hours, and EDC at 6,219.8 hours. These differences warrant further investigation, but they should not be interpreted as a direct ranking of service quality: agencies handle different request types, and some agency groups have relatively small sample sizes.

4. Recorded Due-Date Compliance

A recorded due date was available only for a subset of requests, primarily DSNY/Graffiti requests. Accordingly, this project reports due-date compliance for that subset rather than presenting it as citywide NYC 311 SLA performance.

Due-date subset metric	Result
Requests with recorded due dates	2,224
Validly closed requests analyzed	2,179
Closed by recorded due date	1,468
Closed after recorded due date	711
Completed by due date	67.37%
Completed after due date	32.63%

Correct interpretation: 67.37% of validly closed DSNY/Graffiti requests with recorded due dates were completed by their recorded due date. This is a subset-specific due-date compliance rate, not an official citywide SLA result.

5. Recommendations

- 1. Investigate the long tail.** Review high P90 values and long-running requests to identify recurring causes of extended resolution times.
- 2. Monitor median and P90 together.** Use the median to describe typical performance and P90 to highlight the slowest portion of the workload.
- 3. Plan using geography and problem type.** Compare boroughs and request categories together to target operational reviews more precisely.
- 4. Interpret agency comparisons in context.** Account for request mix, agency responsibilities, and sample size before drawing conclusions.
- 5. Track due-date compliance only where supported.** Use recorded due dates for the eligible subset and avoid implying that this metric covers every NYC 311 request.

6. Limitations & Interpretation

- The analysis covers a selected 50-day period, not the full NYC 311 historical dataset.
- Resolution time measures elapsed time between creation and closure; it does not, by itself, measure service quality or resident satisfaction.
- Missing closed dates and invalid timestamp sequences limit which records can be used for duration metrics, but these records remain in the dataset.
- Due dates were available only for a subset of requests, so the reported compliance rate is not citywide.
- Agency responsibilities and request types differ, and low-volume groups can produce unstable statistics.

Conclusion

This project demonstrates an end-to-end analytical workflow using PostgreSQL and Power BI: exploring public-service data, investigating data-quality issues, defining defensible metrics, identifying demand and geographic patterns, and communicating resolution performance. The central finding is that demand is concentrated in recurring problem categories, while resolution times vary substantially and contain a long tail that is obscured by the average alone.

Prepared by: Muna Mohammed Husen | **Tools:** PostgreSQL, pgAdmin, Power BI, GitHub

Data source: NYC Open Data, 311 Service Requests from 2010 to Present — <https://data.cityofnewyork.us/Social-Services/erm2-nwe9>